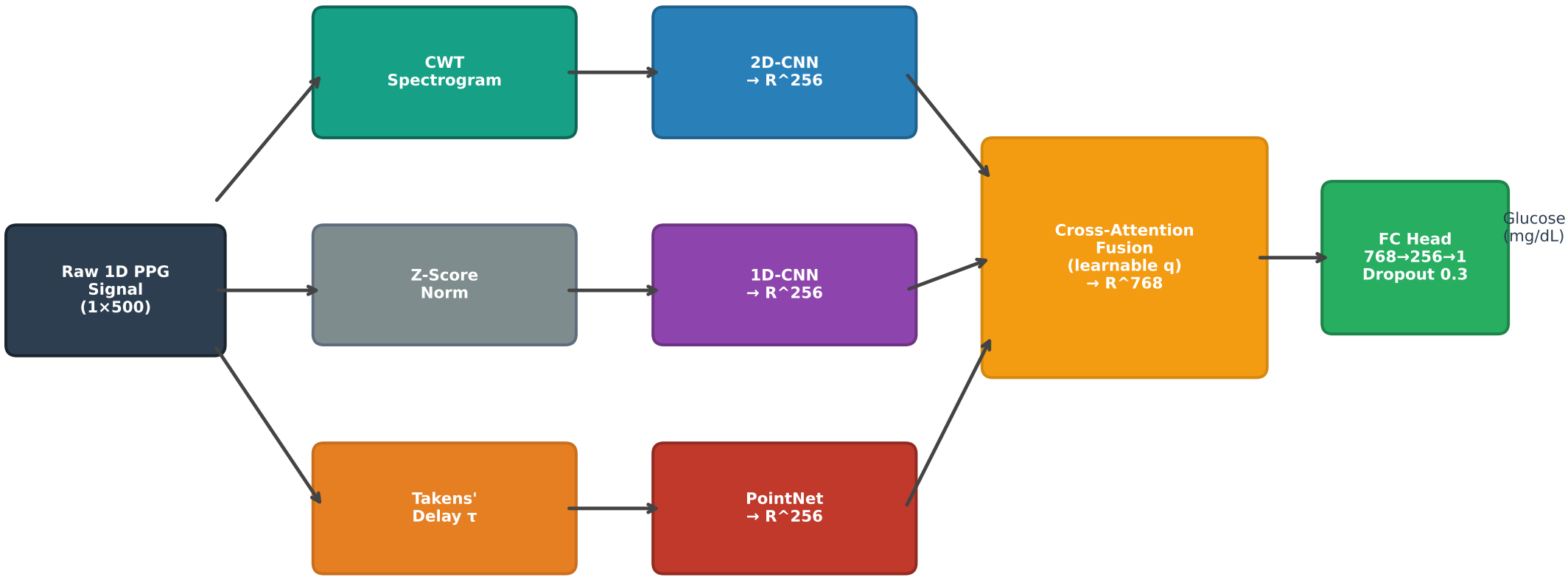
Multimodal 1D+2D+3D Fusion Network Architecture
Raw 1D PPG Signal (1×500)
CWT Spectrogram
Z-Score Norm
Takens' Delay τ
2D-CNN → R^256
1D-CNN → R^256
PointNet → R^256
Cross-Attention Fusion (learnable q) → R^768
FC Head 768→256→1 Dropout 0.3
Glucose (mg/dL)